



Consistent application of periodic boundary conditions in implicit and explicit finite element simulations of damage in composites

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ARTICLE INFO

Keywords:

Periodic Boundary Conditions
Finite elements
Implicit solver
Explicit solver
Micro-mechanics
RVE

ABSTRACT

This paper presents an implementation-dedicated analysis of Periodic Boundary Conditions (PBCs) for Finite Element (FE) models incorporating highly non-linear effects due to plasticity and damage. This research addresses fiber-reinforced composite materials modeled at micro-scale level using a Representative Volume Element (RVE), where its overall mechanical response is obtained via homogenization techniques. For the sake of clearness, a unidirectional ply with randomly distributed fibers RVE model is assumed. PBCs are implemented for implicit and explicit FE solvers, where conformal and non-conformal meshes can be used. The influence of applying PBCs in the reliability of the mechanical response under tension and shear loading is assessed. Furthermore, the Poisson effect and the consistency of damage and fiber debonding propagation through the periodic boundaries are reported as well as their impact on the homogenized results. Likewise, numerical aspects like computational performance and accuracy are evaluated comparing implicit- versus explicit-based solutions.

1. Introduction

Nowadays, the multi-scale approach is well established in the characterization of materials with specific micro-structures. Computational homogenization is used to determine the material behavior on the microscale, then, the homogenized response is applied to macroscale models with concurrent simulations or sequential methods [1]. The micro-structure is usually modeled using a Representative Volume Element (RVE) with periodic boundary conditions (PBCs) [2,3]. The PBCs make possible to represent an infinitely large system using a small domain replicated in the three spatial directions.

PBCs are extensively used in finite element (FE) models, particularly to analyze composite materials on the microscale and mesoscale [4–12]. In the microscale, the small domain is usually a periodic RVE where a random distribution of fibers is embedded in a matrix. The linear elastic, non-linear and damage behavior of the homogenized material are determined based on the elastic, plastic and damage properties of its constituents (fibers, matrix, and interface). The homogenization of the non-linear behavior and damage have been reported in detail in different studies [13,14]. It is important to remark that almost each paper has its own implementation of PBCs [15–21]. All the implementations are based on the same mathematical concept, but

the formulation of the equations and how they are implemented into the FE solver are different. There are no studies which compare the results from different implementations of PBCs.

In addition, it is important to note that there is no direct validation with experimental data for each implementation of PBCs. Some authors have verified that their simulations with PBCs give realistic results. The simulated failure criteria of a unidirectional ply using PBCs has been compared successfully with the experimental one [9,22]. Also, the homogenized stress-strain behavior of a unidirectional ply under different loads has been verified with experimental results having a good correlation at some strain rates [23]. Using PBCs in meso-scale models, such as woven composites, gives a realistic damage evolution when simulations based on PBCs are compared with experiments [24]. Moreover, there is a good correlation between simulated strain fields and the experimental ones for a woven composite [25]. However, none of these studies consider the influence in the results due to the selected implementation of the PBCs, in the same way as a material model is chosen to work with a specific FE solver.

Another important aspect regards the load condition imposed on the RVE to get the homogenized response. Different quasi-static load conditions are usually imposed at the microscale to characterize the material behavior [22,23]. To assess dynamic response, time-dependent

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<https://doi.org/10.1016/j.compositesb.2018.12.023>

Received 8 November 2018; Accepted 7 December 2018

Available online 13 December 2018

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loading conditions like impact or strain rate effects must be included. Although an implicit solver can be used to compute the homogenized dynamic response, an explicit solver is more suitable for time domain events. To our knowledge, there are very few reported investigations using FE explicit analyses with PBCs [26,27]. In [27], Garoz et al. showed that the same analysis performed with an implicit and explicit solver gives different results. During the review process of the present paper, a recent work by Sádaba et al. [28] was brought to the authors' attention, where a user element-based approach is targeted to reduce the differences between both solvers.

The current study focuses on the verification of different implementations of PBCs using both an implicit and an explicit FE solver. Specifically, the homogenized behavior of a unidirectional composite represented by a RVE is compared under different implementations of PBCs. Two different formulations and two implementations of PBCs have been used considering the multiple approaches from previous studies. Although most of the implementations are defined for FE models with conformal meshes, we have included an implementation of PBCs that can be used with non-conformal meshes. One important aspect is the verification of this implementation in implicit and explicit FE solvers.

The formulations and implementations of PBCs in a FE model are presented in section 2 and 3, respectively. Section 4 describes the RVE used in this study. The results of implicit and explicit analyses are discussed in sections 5 and 6, respectively. Finally, the main conclusions are drawn.

2. Formulations of the PBCs in FE

PBCs are formulated in 3D assuming a cuboid unit cell, as shown in Fig. 1 (left). A local coordinate system is located in the vertex A and oriented along the edges of the unit cell, as indicated in Fig. 1 (left). This unit cell is repeated along the three dimensions to mimic a continuous physical body. Displacements at the boundaries between two contiguous unit cells must be connected properly to avoid interpenetration or discontinuities, see Fig. 1 (right). Similarly, the stresses must also respect continuity at the boundaries. These conditions can be satisfied using the first-order computational homogenization [1,17,29]. As a result, the relative displacement between nodes located on a pair of opposite surfaces is given by

$$u_j^{i+} - u_j^{i-} = \varepsilon_{ij}^0 l_i \quad (i, j = x, y, z) \quad (1)$$

where ε_{ij}^0 is the macro-strain tensor of the unit cell, and l_i is the length of the unit cell in i direction between opposite surfaces. l_i is constant given the cuboid geometry of the unit cell. When the macro-strains ε_{ij}^0 are specified, the equations are described by nodal displacement constraints.

If the set of equation (1) is applied to every node on the boundaries of the unit cell, it would result in redundant constraints on the nodes located on the edges. To prevent this redundancy, it is important to define explicitly which equations describe the relative displacement of the nodes on the edges and vertices of the unit cell. Therefore, different

formulations can be selected from this point forward [11,17,19] and are discussed below.

2.1. Relative formulation

In this study, relative and absolute formulations are distinguished. For the sake of simplicity, the difference between both formulations is shown using the equations that describe the displacement of the vertices located on the xOy plane. Using directly the set of equation (1), the displacements of B, D and C are given by

$$u_j^B - u_j^A = \varepsilon_{xj}^0 l_x \quad (j = x, y, z) \quad (2)$$

$$u_j^D - u_j^A = \varepsilon_{yj}^0 l_y \quad (j = x, y, z) \quad (3)$$

$$u_j^C - u_j^B = \varepsilon_{yz}^0 l_z \quad \text{or} \quad u_j^C - u_j^D = \varepsilon_{zx}^0 l_x \quad (j = x, y, z). \quad (4)$$

Only one of equation (4) is needed. Note that the displacement of the node in C is linked with the displacement of the node in B (or D), while the displacement of the nodes in B and D are linked with the displacement of the node in A. Therefore, equations (2)–(4) give the so-called relative formulation of the PBCs.

2.2. Absolute formulation

In the absolute formulation, one vertex and three edges of the unit cell are selected as the local coordinate system. Then, the displacements of the nodes on the other edges and vertices are linked to the nodes located on this *local coordinate system*. Following the example of the vertices located in the xOy plane, the local coordinate system is described by the vertex A as the origin and the edges AB and AD. Equations (2) and (3) already link the displacements of the nodes in B and D to the displacement of the node in A, which is the origin of the local coordinate system. Additionally, the displacements of the node in C can be formulated as

$$u_j^C - u_j^A = \varepsilon_{yj}^0 l_y + \varepsilon_{xj}^0 l_x \quad (j = x, y, z) \quad (5)$$

which is a linear combination of (2) and (4). In this equation, two macro-strains and dimensions are needed to describe the relative displacement between the nodes.

The absolute formulation for PBCs is used if the displacements of the nodes on vertices and edges are directly linked with the nodes laying on the *local coordinate system*. Otherwise, the relative formulation is used. Although both formulations give an equivalent system of equations, the implementation into the finite element solver can lead to some differences which are discussed in sections 5 and 6 for implicit and explicit solvers, respectively.

3. Implementation of PBCs in FE

As described before, the geometry is a cuboid unit cell on which defined macro-strains are applied. Because the dimensions l_i of the unit cell are fixed, the term $\varepsilon_{ij}^0 l_i$ ($i, j = x, y, z$) can be described as the displacement of a dummy node $\hat{u}_j^i$. For each direction i that links nodes in opposite surfaces, there is a dummy node with three displacement components j . In this way, linear constraints of the displacement between opposite nodes can be easily implemented in a FE solver [17,19]. The full macro-strain tensor can be defined by the displacement components of the dummy nodes.

The full set of equations to apply PBCs on the cuboid unit cell represented in Fig. 1 (left) is written in Tables 1 and 2 for the relative and absolute formulation, respectively. They are sorted in three groups: (i) internal nodes of the surfaces; (ii) nodes at the edges except vertices; and (iii) vertices. The notation of the surfaces and edges follows the nomenclature of the vertices of each entity, see Fig. 1. Note that only 3 equations are necessary for the internal nodes of the surfaces, 9 equations for the nodes at the edges except vertices, and 7 equations for the vertices.

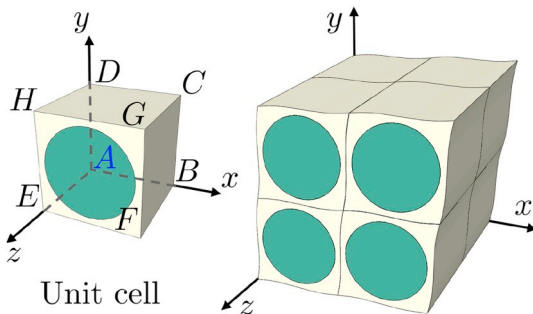


Fig. 1. Unit cell and periodic array.

Table 1
PBCs equations for the relative formulation ($j = x, y, z$).

Surfaces	Edges	Vertices
$u_j^{DHGC} - u_j^{AEFB} = \hat{u}_j^y$	$u_j^{DH} - u_j^{AE} = \hat{u}_j^y$	$u_j^D - u_j^A = \hat{u}_j^y$
	$u_j^{HG} - u_j^{EF} = \hat{u}_j^y$	$u_j^H - u_j^E = \hat{u}_j^y$
	$u_j^{CG} - u_j^{BF} = \hat{u}_j^y$	$u_j^C - u_j^B = \hat{u}_j^y$
	$u_j^{DC} - u_j^{AB} = \hat{u}_j^y$	$u_j^D - u_j^B = \hat{u}_j^y$
$u_j^{BCGF} - u_j^{ADHE} = \hat{u}_j^x$	$u_j^{BC} - u_j^{AD} = \hat{u}_j^x$	$u_j^B - u_j^A = \hat{u}_j^x$
	$u_j^{FG} - u_j^{EH} = \hat{u}_j^x$	$u_j^F - u_j^E = \hat{u}_j^x$
	$u_j^{BF} - u_j^{AE} = \hat{u}_j^x$	
$u_j^{EFGH} - u_j^{ABCD} = \hat{u}_j^z$	$u_j^{EF} - u_j^{AB} = \hat{u}_j^z$	$u_j^E - u_j^A = \hat{u}_j^z$
	$u_j^{FH} - u_j^{AD} = \hat{u}_j^z$	

Table 2
PBCs equations for the absolute formulation ($j = x, y, z$).

Surfaces	Edges	Vertices
$u_j^{DHGC} - u_j^{AEFB} = \hat{u}_j^y$	$u_j^{BF} - u_j^{AE} = \hat{u}_j^x$	$u_j^B - u_j^A = \hat{u}_j^x$
	$u_j^{DH} - u_j^{AE} = \hat{u}_j^y$	$u_j^D - u_j^A = \hat{u}_j^y$
	$u_j^{CG} - u_j^{AE} = \hat{u}_j^x + \hat{u}_j^y$	$u_j^C - u_j^A = \hat{u}_j^x + \hat{u}_j^y$
$u_j^{BCGF} - u_j^{ADHE} = \hat{u}_j^x$	$u_j^{EF} - u_j^{AB} = \hat{u}_j^z$	$u_j^E - u_j^A = \hat{u}_j^z$
	$u_j^{DC} - u_j^{AB} = \hat{u}_j^y$	$u_j^D - u_j^A = \hat{u}_j^y + \hat{u}_j^z$
	$u_j^{HG} - u_j^{AB} = \hat{u}_j^y + \hat{u}_j^z$	$u_j^H - u_j^A = \hat{u}_j^y + \hat{u}_j^z$
$u_j^{EFGH} - u_j^{ABCD} = \hat{u}_j^z$	$u_j^{BC} - u_j^{AD} = \hat{u}_j^x$	$u_j^C - u_j^A = \hat{u}_j^x + \hat{u}_j^y + \hat{u}_j^z$
	$u_j^{FG} - u_j^{AD} = \hat{u}_j^z$	
	$u_j^{FH} - u_j^{AD} = \hat{u}_j^x + \hat{u}_j^z$	

3.1. Dummy nodes

The dummy nodes are usually introduced as reference points which are not attached to any other part of the model, see Fig. 2. These reference points make possible to set easily any kind of boundary condition. Then, the model can be reused for different load conditions changing only the displacement of these reference points. In addition, because the dummy node is a point of the model, different boundary conditions can be applied to it, such as concentrated forces or fixed displacements in one or several components. When a displacement component of the dummy node is set free, this displacement is part of the array of unknown variables and, therefore, it is computed by the FE solver under free stress condition. As a result, the Poisson effect in different directions is easily permitted.

The position of the reference points is not relevant because the link between these points and the nodes of the unit cell is only established

with the kinematic linear constraints. Note that they are not part of any solid element. This approach is intuitive and useful if the simulations are carried out under static loads in an implicit solver. But, in dynamic simulations using explicit solvers, every node and point should have a mass and it is not trivial to determine the mass of the dummy nodes. To avoid this problem and simplify the implementation of PBCs, we have used the nodes at the vertices B , D , and E (see Fig. 1 left) as dummy nodes of the linear constraints. Therefore, the implementation of the set of equations as linear constraints is done as before except for the fact that the equations describing the relative displacement of these nodes at the vertices must be removed. Finally, because the nodes belong to a solid element, they have an associated mass.

3.2. Node-to-node and surface-coupling implementations

Regardless of using the relative or the absolute formulation, the set of equation (1) is written using a node-to-node fashion. This implementation is called node-to-node coupling and requires a conformal mesh, where each node on one surface has its counterpart node on the opposite surface.

When the mesh is non-conformal, the nodes on one surface of the unit cell do not necessarily have counterpart nodes on the opposite surface. Therefore, another implementation is needed. There are some approaches that overcome this problem: (i) copying the nodes from one surface to the opposite one [17]; (ii) making polynomial interpolation functions at the surfaces [30]; (iii) creating new nodes at both surfaces which are counterpart nodes [31]. In the current paper, we have adopted the first approach (i) described by Al Kassem [17] which is easy to implement in a finite element solver.

The implementation has been done in the cuboid unit cell. Three pairs of surfaces can be described along each axis of the local coordinate system. Then, the surfaces perpendicular to the x axis are noted as $(x-)$ and $(x+)$. The same notation is applied for the surfaces perpendicular to the y and z axes. First, the nodes from the surface $(x-)$ are copied to the opposite surface $(x+)$ as reference points, see copied nodes $(x-)$ in Fig. 2. Then, the set of equation (1) can be applied between the original nodes of the surface $(x-)$ and the copied nodes. Finally, the copied nodes are coupled with the original nodes of the surface $(x+)$. The same procedure is done in the other two directions y and z . We call this implementation of the PBCs surface-coupling because the copied nodes from $(x-)$ are tied to the corresponding surface $(x+)$. It is noteworthy that this implementation can be used with conformal and non-conformal meshes.

The surface-coupling is easy to implement with the relative formulation described in Table 1, because all the copied nodes are grouped as surfaces which include nodes on the edges and vertices. Then, the surface of copied nodes can be directly tied to the surface constructed from the original nodes of the surface (see the example with red points in Fig. 2, where the surfaces of copied nodes $(x-)$ and nodes $(x+)$ contain internal nodes of the surface and also nodes at the edges). In the case of the absolute formulation (see Table 2), the nodes of the vertices and edges must be separated. Then, the nodes inside the surfaces are tied as before using surface groups, but the copied nodes from the edges are only tied to the nodes located on the edge except the vertices. Finally, the vertices are coupled directly without any copied nodes.

The described implementations of the PBCs equations can be used in any unit cell with a cuboid shape. For example, the unit cell can be a statistically Representative Volume Element (RVE) which conforms the geometry of unidirectional fibers embedded in a matrix, or a Representative Unit Cell (RUC) that describes the pattern of a textile composite. In the following section, PBCs equations are applied to a periodic RVE with a random distribution of unidirectional fibers in order to study the damage behavior under different loads.

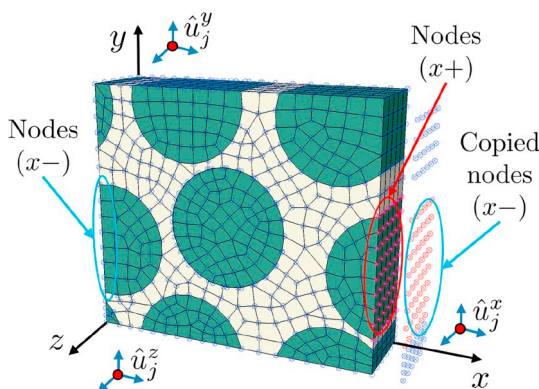


Fig. 2. Unit cell with the surface-coupling implementation of PBCs.

4. Microscale model with PBCs

A microscale RVE model is used to represent a unidirectional fiber composite where its components (fibers, matrix, and the interface) can be damaged. The homogenized damage behavior is studied under different loads assuming PBCs.

4.1. Geometry

The RVE model consists of a periodic cuboid geometry that contains a random distribution of fibers parallel to the z direction, see Fig. 2. All the fibers have the same radius, $r_f = 2.6 \mu\text{m}$ [32]. The number of fibers and the volume fraction are fixed to $n_f = 4$ and $v_f = 0.6$, respectively. The dimensions of the section transverse to the fibers, l_x and l_y , are calculated from n_f , r_f and v_f assuming an initial hexagonal package distribution of fibers. Therefore, $l_x = 12.79 \mu\text{m}$ and $l_y = 11.07 \mu\text{m}$. The depth of the RVE will be set as a function of the element size and the number of elements along the z direction. The random distribution of the fibers was computed with a collision model, for more details see Ref. [33]. The authors are aware that for statistically relevant results, the simulated behavior must be verified by increasing the number of fibers for each load case. This is out of the scope of this work which is focused on the formulation and implementation of PBCs.

4.2. Materials

Different constitutive models are used to describe the behavior of each constituent, fiber and matrix, and the interface between both. The carbon fibers have a linear-elastic transversely isotropic material model. Table 3 shows the material properties. For the sake of simplicity, fiber failure is not considered in this study. The matrix is modeled using an isotropic plastic damage model based on [34,35] which is included in Abaqus [36]. This model has been used to represent a realistic behavior of polymers [9,37,38]. The material parameters of the matrix have been set to fit the experimental data from Ref. [39], see Fig. 2 in Ref. [33]. More detail about the description of the material parameters and their assumed values is reported in Ref. [33]. Table 4 shows: the isotropic elastic properties for tension and compression before yield; the maximum strain for tension, compression and shear; and the density of the matrix. Finally, the interface between both constituents, fiber and matrix, is modeled using a cohesive surface contact formulation [36], which is based on the cohesive zone model [40,41] under implicit and explicit solvers. Table 5 shows the material parameters describing the cohesive traction-separation law under tension and shear. Under compression, hard contact between both constituents is assumed.

4.3. Implementations of PBCs

Table 6 shows the four scenarios studied in this work. Two scenarios use the node-to-node coupling: one with the relative and the other with the absolute formulation. The other two scenarios use the surface-coupling, again one with the relative and the other with the absolute formulation.

The scenarios with the node-to-node coupling are only applicable when the microscale model has a conformal mesh, while the surface-coupling can be applied for models with a conformal or a non-

Table 3

Properties of carbon fibers [32]. E_{11} , E_{22} = longitudinal and transverse Young's modulus, respectively; ν_{12} = Poisson's ratio; G_{12} , G_{23} = in-plane and transverse shear modulus, respectively; ρ = density.

E_{11}	E_{22}	ν_{12}	G_{12}	G_{23}	ρ
(GPa)	(GPa)	(–)	(GPa)	(GPa)	(Kg/m3)
276	15	0.2	15	7	1780

Table 4

Properties of epoxy matrix [39]. E = Young's modulus; ν = Poisson's ratio; ϵ_f^T , ϵ_f^C , γ_f^S = maximum strain at failure for tension, compression, and shear respectively; ρ = density.

E	ν	ϵ_f^T	ϵ_f^C	γ_f^S	ρ
(GPa)	(–)	(–)	(–)	(–)	(Kg/m3)
3.76	0.39	0.058	0.36	0.58	1300

Table 5

Properties of the interface [32]. τ_n , τ_t = normal and transverse interface strength; G_{Ic} , G_{IIc} = Mode I and Mode II interface critical energy release rate; η = mixed mode interaction parameter (power law fracture criterion [35]).

τ_n	τ_t	G_{Ic}	G_{IIc}	η
(MPa)	(MPa)	(N/mm)	(N/mm)	(–)
50	75	0.002	0.006	1.45

Table 6

Studied scenarios for different implementation of PBCs according to the coupling and formulation.

Scenario	Coupling	Formulation
N2N-REL	Node-to-node	Relative
N2N-ABS	Node-to-node	Absolute
SC-REL	Surface-coupling	Relative
SC-ABS	Surface-coupling	Absolute

conformal mesh. In order to compare the results of the different scenarios, microscale models with a conformal mesh are studied in sections 5 and 6.

Scenarios with surface-coupling implementations make debonding possible of the interfaces with nodes on the surfaces of the RVE where PBCs apply. To do that, the copied nodes ($x-$) are grouped in several surfaces which are separated by the interfaces between fibers and matrix. Then, each surface of copied nodes ($x-$) is separately tied with its corresponding surface of nodes ($x+$), see Fig. 2. This implementation has been done independently of the PBCs formulation.

4.4. Load conditions

Load conditions are imposed through the dummy nodes as discussed in section 3. Displacements or concentrated forces along the three directions can be applied to each dummy node. Therefore, different load conditions can be studied, such as uniaxial tension or shear under displacement or load control. In addition, the load conditions can be a combination of displacements and concentrated forces applied at the different components of the dummy nodes.

In sections 5 and 6, static and dynamic loads are studied, respectively. The model under static loads makes use of an implicit solver, therefore only the total displacement or force applied on the dummy nodes is imposed. However, the model under dynamic loads needs an explicit solver and then, the displacement or force must be a function of time.

4.5. Homogenized properties

The results obtained from the simulation of the microscale model are the local stress and strain fields inside the unit cell. The response of the unidirectional composite is obtained from the homogenized stress (σ_{ij}) versus the homogenized strain (ϵ_{ij}). The homogenized stress can be calculated for each time step using

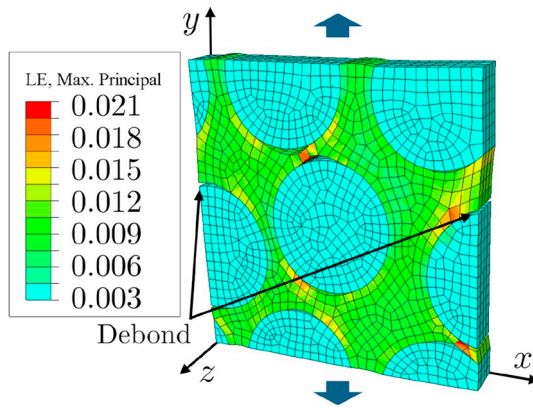


Fig. 3. Debonding and maximum principal logarithmic strain field of the RVE under uniaxial load transverse to the fiber using the surface-coupling implementation. A deformation scale factor of 10 has been applied to highlight the deformations.

$$\sigma_{ij} = \frac{1}{V} \int_V \sigma_{ij}^{loc} dV \quad (i, j = x, y, z) \quad (6)$$

where σ_{ij}^{loc} is the local stress tensor, V is the volume of the RVE. On the other hand, the homogenized strain is equal to the macro-strain of the dummy nodes as a result of the first-order computational homogenization, $\varepsilon_{ij} = \varepsilon_{ij}^0$. It is noteworthy that the homogenized strains (or macro-strains) take into account non-linear displacements inside the unit cell like the opening of the interfaces when there is debonding.

It is important to remember that the formulation of PBCs introduces some constraints and limitations related to the periodicity of the results. It means that the damage created in the RVE represents a periodic damage affecting the whole represented material and not a local defect. Moreover, edge effects due to the geometry of the laminate or macro-structure are out of the scope of the present study.

5. Results using PBCs under static loads (implicit solver)

In this section, the results of a finite element model with PBCs is studied under static loads using the implicit solver Abaqus/Standard [36]. As an illustrative case, a periodic RVE with 4 fibers has been used in the simulations described below, see Fig. 3. The length in the fiber direction l_z varies as a function of the global element size of $0.4 \mu\text{m}$, from 1 to 8 times this size, $3.2 \mu\text{m}$.

Matrix and fibers are described by two solid parts. Each part is discretized with hexahedral and/or wedge elements to generate a conformal mesh usable for any of the scenarios expressed in Table 6. Two conformal meshes have been studied: (i) a hexahedral-dominated mesh where almost all the solid elements are 8-node-hexahedral elements with the exception of a few 6-node-wedge elements; and (ii) a mesh where all elements are wedge. Table 7 shows the number of solid elements contained in each mesh when there is one layer of elements through the depth ($l_z = 0.4 \mu\text{m}$) or four layers ($l_z = 1.6 \mu\text{m}$). Additionally, the influence of using full integration (8 integration points in hexahedral elements, and 2 integration points in wedge elements) and

Table 7

Number of hexahedral and wedge elements contained in each conformal mesh with one layer of elements through the depth ($l_z = 0.4 \mu\text{m}$) or four layers ($l_z = 1.6 \mu\text{m}$).

Mesh	l_z	Hexahedral elements	Wedge elements
Hexahedral-dominated	0.4	1124	40
Wedge mesh	0.4	0	2036
Hexahedral-dominated	1.6	4496	160
Wedge mesh	1.6	0	8144

reduced integration (one integration point in hexahedral elements) has been studied. The material properties of the fibers and the matrix described in section 4.2 have been applied to the respective solid parts. In addition, cohesive surface elements are included between the solid parts with the cohesive law properties from Table 5. It has been confirmed that there is mesh convergence of the described model with a global element size of $0.4 \mu\text{m}$. This size also agrees with the requirements for the critical cohesive length proposed by Turon et al. [42].

5.1. Damage propagation through the boundaries

In the current study, periodicity has been assumed in the linear and non-linear behavior. Therefore, the damage must propagate through the boundaries of the RVE. The implementation of the PBCs must allow the damage propagation in the matrix and the debond at the interfaces. The authors recognize that this is a strong hypothesis and it is only realistic under certain loads and material properties when the damage in the material is diffuse, for example when a shear load is applied [32]. Nevertheless, the implementations of the PBCs have been tested for two different loads, (i) transverse to the fiber direction and (ii) in-plane shear, in order to check the suitability as well as possible drawbacks.

Periodic damage propagation means that the damage in an element laying on the boundary surface of the RVE will continue in the opposite surface. When the node-to-node coupling is applied to a conformal mesh, the displacements of counterpart nodes between opposite surfaces are directly linked. Therefore, for the matrix modeled with a continuum damage model, the damage of the solid element is transferred to the element in the opposite surface. In addition, the debond of a cohesive surface between fibers and matrix will also be transferred to the cohesive surface on the opposite surface.

For the case of surface-coupling with a conformal or non-conformal mesh, the damage propagation must be handled with care because the displacements from one surface ($x-$) are transferred to the opposite surface ($x+$) by means of the copied nodes, see Fig. 2. It is important to remark that the surface-coupling implementation presented in this paper makes it possible the propagation of a debond using cohesive surfaces between opposite surfaces of the RVE. However, the propagation of the continuum damage of a solid element can be restricted due to the fact that the surface-to-surface constraints are based on a master-slave relation [16]. The displacements of a slave node are described by the displacements of the closest master nodes using element shape functions. In our implementation, the copied nodes are the masters ones. For example, if a slave node from ($x+$) lays on the middle of the vertical position of two copied nodes from ($x-$), the constraints introduce a geometrical restriction in the vertical displacement of these copied nodes. The authors are aware that this possible restriction can affect the failure of the composite when large displacements and strains ($>10\%$) are expected in specific areas of the matrix. This failure is out of the scope of the present study which is focused on the initial failure where matrix plasticity and debonds at the interfaces are initiated and propagated.

Damage propagation has been studied with the described micro-mechanical model using node-to-node and surface-coupling implementations. It has been verified that both implementations do not affect the initial failure of the unidirectional composite represented by the RVE. The initial failure is represented by the initiation and propagation of several debonds [43]. Fig. 3 shows the debonds and the maximum principal logarithmic strain field of the RVE under uniaxial tension transverse to the fiber using the surface-coupling implementation. This implementation gives the same nodal displacements as the node-to-node implementation (not shown in the figure). It can be observed that several debonds have propagated across opposite surfaces while the maximum principal strain is below the maximum strain at failure for tension, $\varepsilon_f^T = 0.058$.

5.2. Poisson effect

In this section, two cases are compared to each other: one where the Poisson effect is allowed and one where it is restricted in the RVE model. Higher stresses are expected when the Poisson effect is restricted and, as a consequence, premature failure. The material characterization must be done as in real experiments where the Poisson effect is allowed.

Four load conditions are studied, two uniaxial and two shear load cases. The periodic RVE with 4 fibers and depth of $1.6 \mu\text{m}$ has been discretized in a hexahedral dominated mesh with full integration elements. The studied load cases of this section have been only performed with the N2N-ABS implementation of PBCs. The results with different implementations are discussed in section 5.5.

The uniaxial load conditions are called uniaxial tensile stress and uniaxial tensile strain. In the uniaxial tensile stress case, only one axial component of stress is positive. Using the displacement control on the dummy nodes, the uniaxial tensile stress along y is imposed through a fixed displacement u_0 in the axial direction, $\hat{u}_y^y = u_0$, while the axial components of the other dummy nodes, $\hat{u}_x^x$ and $\hat{u}_z^z$, are unconstrained. Additionally, the other components of the three dummy nodes related to shear are also free. A free displacement component of a dummy node means that the reaction force of that component is zero. As a result, the RVE under uniaxial tensile stress will suffer contraction due to the Poisson effect and the only nonzero reaction force will be the one in the axial direction of the y axis, i.e. the direction of the imposed displacement. It is important to remark that this load condition describes a usual experiment for material characterization under tension.

On the other hand, when uniaxial tensile strain is imposed, only one axial strain is nonzero. In this case, the uniaxial tensile strain along y is imposed directly with a fixed displacement u_0 , $\hat{u}_y^y = u_0$. The axial components of the other dummy nodes are set to zero, $\hat{u}_x^x = \hat{u}_z^z = 0$. Hence, the Poisson effect is restricted and the RVE is not allowed any contraction in the x or z direction. Reaction forces along the axial components of each dummy node will be generated due to the load condition. Contrary to the uniaxial tensile stress, the uniaxial tensile strain cannot be imposed easily in a real experiment.

The axial stress in the y direction (σ_{yy}) for the uniaxial stress and uniaxial strain load conditions for a certain displacement $\hat{u}_y^y$ is display in Fig. 4(a) and (b), respectively. It can be observed that the maximum stress in the y direction before any damage is higher in the uniaxial strain load condition due to the restriction of the Poisson effect. Hence, debonding and matrix damage are expected earlier in the uniaxial strain load condition (i.e. restricted Poisson) than in the uniaxial stress case (i.e. free Poisson). This behavior is shown in Fig. 5 where the curve for uniaxial restricted Poisson exhibits a higher stiffness and a lower

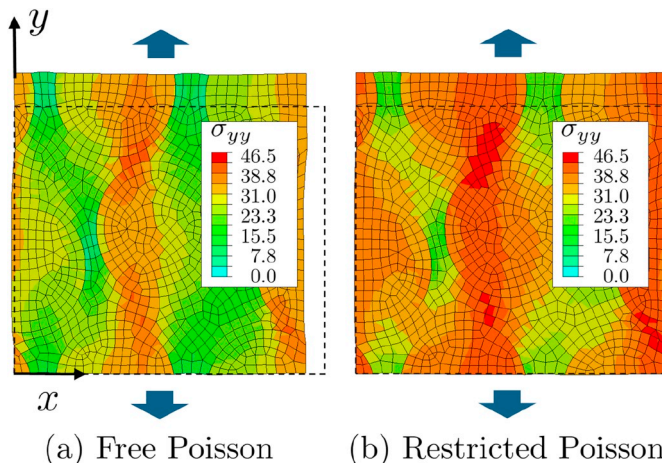


Fig. 4. Vertical stress field (σ_{yy}) for two different load cases: (a) uniaxial tensile stress with free Poisson; and (b) uniaxial tensile strain with restricted Poisson.

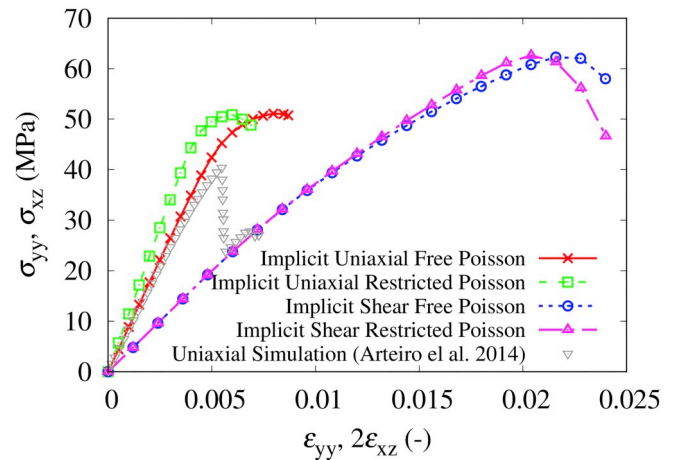


Fig. 5. Homogenized stress-strain responses for four different load cases: uniaxial tension with free Poisson effect, uniaxial tension with restricted Poisson effect, shear load with free Poisson effect, and shear load with restricted Poisson effect. Hexahedral dominated mesh with full integration elements and depth $l_z = 1.6$. Uniaxial simulation done by Arteiro et al. [32]

strain at maximum stress than the case for uniaxial free Poisson. It is thus important to reproduce a realistic load case, i.e the uniaxial stress condition where the Poisson effect is allowed, to characterize the damage behavior of a material under tension.

The current implementation of PBCs makes it possible to impose a shear load with the Poisson effect restricted or not. For example, pure shear in the xOz plane is imposed with a fixed displacement on the lateral components of the dummy nodes related to the x and z axes, $\hat{u}_x^x = \hat{u}_z^z = u_0$. The rest of the components of the dummy nodes are unconstrained to allow the Poisson effect. It is true that for a linear isotropic material under pure shear, no Poisson effect is expected. However, this is not the case for a material with a non-linear orthotropic behavior. On the other hand, simple shear in the xOz plane is applied, fixing the displacements on the lateral components $\hat{u}_x^x = \hat{u}_z^z = u_0$ and setting the axial components of the three dummy nodes to zero, $\hat{u}_y^y = \hat{u}_x^y = \hat{u}_z^y = 0$. In this latter case, the Poisson effect is restricted.

Fig. 5 also shows the material response under a shear load with free or restricted Poisson. The differences are visible at the end of the curves in the non-linear region where debonding and plasticity of the matrix are relevant. As in the uniaxial load case, the maximum stress occurs at a lower strain when Poisson effect is restricted. Hence, also for shear, restricting Poisson contraction introduces higher local stresses that lead to premature damage.

Finally, for the sake of verification, the simulated behavior of an embedded 90° ply for sublaminae with 0° outer plies under uniaxial tension [32] has been included in Fig. 5. Although the material model of the matrix and interface are not exactly the same implementation as the ones presented in this paper, the input parameters for fiber, matrix, and interface are the same. As expected, the uniaxial tensile case (i.e. free Poisson) imposed with PBCs is in line with the simulated behavior of the embedded model. There are some differences due to the geometry: (i) the initial slope of the elastic behavior is smaller in the embedded model because it has a smaller volume fraction ($vf = 0.591$); (ii) the sudden fall in the stress due to the first transverse matrix crack cannot be modeled using the RVE with 4 fibers which only can account for diffuse damage with smooth degradation of the stress.

5.3. Depth of RVE

The results of RVEs with different depths ($l_z = 0.4, 1.6$ and $3.2 \mu\text{m}$) have been compared under different load cases. Hexahedral dominated mesh with the N2N-ABS implementation has been used in all cases.

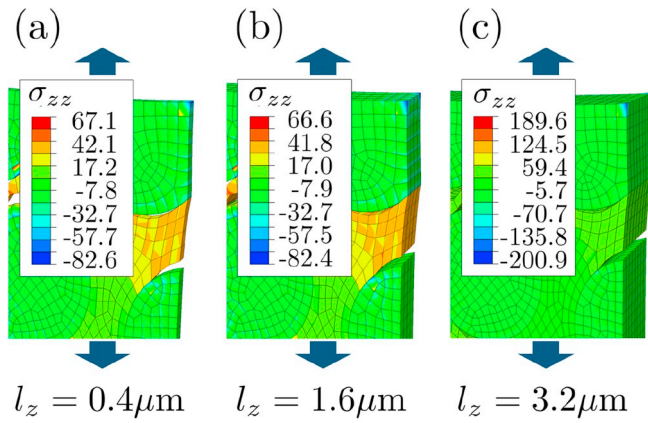


Fig. 6. Stress field in fiber direction σ_{zz} of the three RVEs with different depth l_z when a vertical macro-strain $\varepsilon_{yy}^0 = 0.007$ is applied. A deformation scale factor of 30 has been applied to highlight deformations and debonds.

First, the uniaxial tensile stress (free Poisson) is applied in the vertical direction y . There are no significant differences between the homogenized stress-strain responses in the linear and non-linear regime. These results agree with the study of Melro et al. [44] who showed that the depth of the RVE does not affect the homogenized transverse elastic properties. However, at the end of the simulations, debonding and the plastic strain of the matrix are notable. Due to convergence issues of the material models, each simulation stops at a different final strain ε_{yy} . Additionally, high local stresses are found in the elements located at the vertices of the RVE. Fig. 6 shows the stress field in fiber direction σ_{zz} of the three RVEs with different depths when a vertical macro-strain of 0.007 is applied.

Two other load cases have been checked using the same RVEs with different depths. When uniaxial tensile strain (restricted Poisson) is applied, the same behavior as in the case of uniaxial tensile stress (free Poisson) is observed. Convergence issues with high local stresses in the elements located at the vertices are obtained at the end of the simulations. On the other hand, when shear load is applied to the same geometry and material models using PBCs, there are no convergence issues and no high local stresses in the elements located at the vertices.

High local stresses and convergence issues are a consequence of the combination of the geometry, material models, load case and PBCs. Because the uniaxial tensile load case is the most relevant, additional simulations have been done, increasing the number of fibers and modifying the material behavior of the fiber, matrix, and interface. First, the results using an RVE with 30 fibers have been compared showing the same homogenized behavior than the RVE with 4 fibers. High local stresses and convergence issues are also found in the RVE with 30 fibers. Secondly, simulations using the RVE with 4 fibers have

been performed modifying the material behaviors. When a linear isotropic fiber, i.e. glass fiber, is modeled, the high local stresses and convergence issues are still present. In this case, the relevant stresses on the elements at the vertices are σ_{yy} . When the plastic matrix behavior is replaced for a linear elastic one, the results show also high stresses at the element located at the vertices with convergence issues. Finally, when high interface strength has been imposed to avoid debonding, the simulations do not show high local stresses at the elements of the vertices and there are no convergence issues. In addition, these latter results are independent of the fiber or matrix material models.

As a conclusion, the cohesive surfaces that model the brittle debonding between fibers and matrix combined with the PBCs can lead to artificial stresses in located elements. These stresses can be more relevant in some load cases like the uniaxial stress one. Results with these high artificial stresses must be handled with care and they must be omitted from the homogenized behavior.

5.4. Element types

Three different solid element types have been used to discretize the studied RVE with 4 fibers: hexahedral elements with full integration (C3D8) [36]; hexahedral elements with reduced integration (C3D8R); and wedge elements with full integration (C3D6). The results are analyzed and compared in this section.

As reported before, the RVE with 4 fibers and $l_z = 1.6 \mu\text{m}$ has been discretized with a hexahedral dominated mesh where almost all the solid elements are hexahedral (4496) except a few wedge elements (160) needed to reach a conformal mesh, see Table 7. Hexahedral elements with full or reduced integration can be chosen, hence a comparison between both can be performed. Note that the wedge elements have only full integration in Abaqus/Standard [36], but their contribution in the case of a reduced integration mesh is limited. A second discretization using only wedge elements (8144) has been done in order to compare it with the hexahedral dominated mesh with full integration. In all the cases, the meshes are conformal.

Fig. 7 shows the vertical stress field σ_{yy} when a vertical macro-strain of $\varepsilon_{yy}^0 = 0.0065$ is applied under uniaxial tensile load with free Poisson for the three cases described above. At this macro-strain, there are debonding between several interfaces. It can be observed that the average stress fields are identical for the wedge mesh (a) and the hexahedral dominated mesh (b) when both of them have full integration elements. This agreement can be seen along the whole simulation. Indeed, the wedge mesh has the same convergence issues as the hexahedral dominated mesh when the uniaxial load is applied. At the end of the simulations, there are high local stresses in the elements located at the vertices. As shown in section 5.3, the simulations of RVEs with different depths stop at different final strains due to convergence issues.

A different result shows up in the hexahedral mesh with reduced integration elements. Fig. 7 (c) shows that the solid elements of the

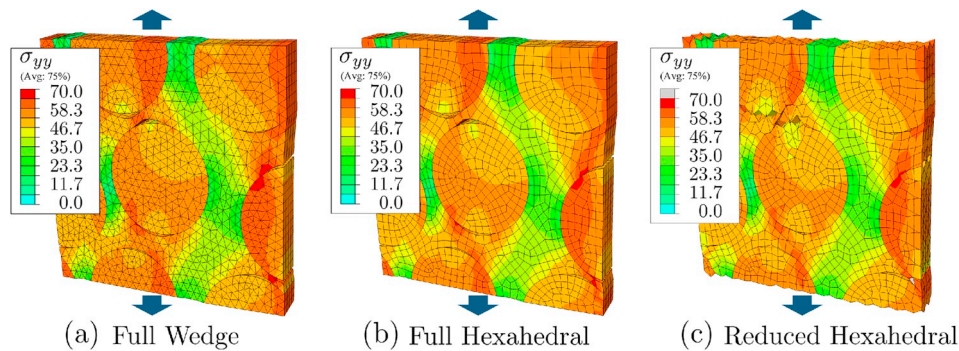


Fig. 7. Vertical stress field σ_{yy} of three RVEs when a vertical macro-strain $\varepsilon_{yy}^0 = 0.0065$ is applied: (a) Wedge mesh with full integration; (b) hexahedral dominated mesh with full integration; (c) hexahedral dominated mesh with reduced integration. A deformation scale factor of 20 has been applied to highlight deformations and debonds.

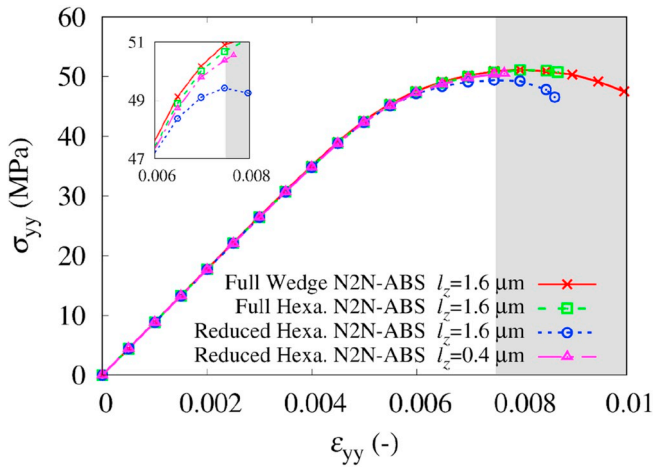


Fig. 8. Homogenized stress-strain response for three different meshes of the RVE $l_z = 1.6 \mu\text{m}$ using full integration wedge elements, full integration hexahedral elements, and reduced integration hexahedral elements. In addition, the response of the RVE $l_z = 0.4 \mu\text{m}$ with reduced integration hexahedral elements is included. Implicit solver with N2N-ABS implementations under uniaxial tensile load with free Poisson.

mesh suffer hourglass modes. Note that a scaling factor of 20 has been applied to see the deformations. Hourglassing means that solid elements deform under non-physical modes without stress contribution. Therefore, the results are not realistic and must be discarded. The hourglassing is triggered by the combination of the geometry, material models, load conditions and PBCs. Specifically, in the simulated case with the vertical uniaxial load, the hourglassing becomes relevant during the propagation of the debond.

Fig. 8 compares the homogenized stress-strain of the three different meshes for the RVE with $l_z = 1.6 \mu\text{m}$. Results with homogenized strains beyond 0.75% must be discarded due to convergence issues reported in section 5.3, see the gray area in Fig. 8. The wedge and hexahedral dominated meshes with full integration elements give a similar response with differences in stress below 0.1 MPa. The hexahedral mesh with reduced integration elements ($l_z = 1.6 \mu\text{m}$) underestimates the stress when debond propagates. This response is due to the hourglass mode. The hourglassing can be limited using a RVE with only one layer of elements along the depth, see the results for $l_z = 0.4 \mu\text{m}$ included in Fig. 8. Then, the homogenized stress-strain response is closer to the results using wedge or hexahedral meshes with full integration elements.

Finally, the RVEs with the three different meshes have been studied under uniaxial tensile strain (restricted Poisson) and in-plane shear loading. The convergence issues are again present when the uniaxial tensile strain is applied. In that case, the results show the same behavior as the ones reported for uniaxial tensile stress (free Poisson). Under in-plane shear load, the stress fields and homogenized results are practically identical for the three meshes. There are no convergence issues and the hourglassing is negligible in the reduced integration mesh.

5.5. Results with different implementations of PBCs

Although the 4 scenarios described in Table 6 must be equivalent, there can be differences between them due to the implementation in a specific solver. In this section, the results of the simulations computed in Abaqus/Standard [36] are compared.

In order to compare the 4 scenarios, the RVE with 4 fibers and a depth of $1.6 \mu\text{m}$ has been discretized using a hexahedral dominated mesh with full integration elements. Fig. 9 shows the homogenized stress-strain response of the 4 scenarios when a static uniaxial tensile stress (free Poisson) is applied in the vertical direction y. No significant differences can be observed between the 4 scenarios in the linear

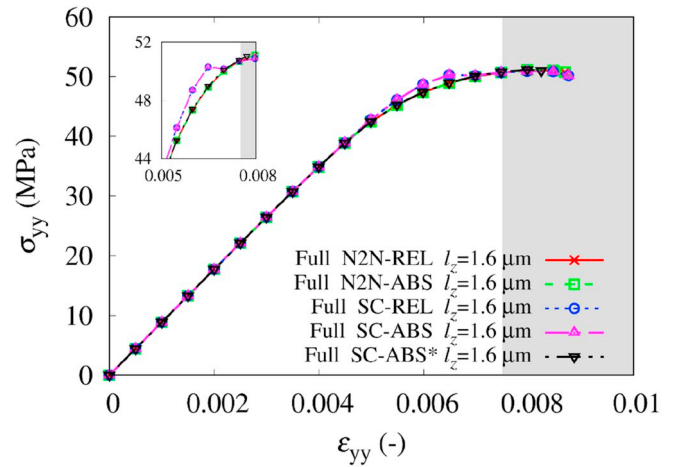


Fig. 9. Homogenized stress-strain response for the four different implementations under uniaxial load with free Poisson. RVE with $l_z = 1.6 \mu\text{m}$ discretized with full integration hexahedral dominated mesh. SC-ABS* surface-coupling implementation with user defined tolerance (10^{-6}).

behavior. On the contrary, in the non-linear behavior, there is a small difference between the node-to-node and surface-coupling implementations, compare curves N2N to SC in Fig. 9. These differences appear during the propagation of debonding ($\epsilon_{yy} > 0.0055$), see inset of Fig. 9. The differences between the node-to-node and surface coupling implementations are due to the solver tolerance limit based on the average force. In the surface-coupling implementation, new surfaces of nodes are created in the model and the solver changes the tolerance limit based on another criterion. To obtain the same results using both node-to-node and surface-coupling approaches the tolerance limit based on the average force must be defined by the user. Using a value of 10^{-6} , the stress-strain behavior of the surface-coupling implementations overlaps the results of the node-to-node implementations, see the curve SC-ABS* in Fig. 9.

It is worth mentioning that the same results were obtained for different depths (0.4, 1.6 and $3.2 \mu\text{m}$). As commented in section 5.3, there are convergence issues at the end of the simulations. Hence, the results with homogenized strains beyond 0.75% must be discarded, see the gray area in Fig. 9. Similarly, the same response and conclusions were drawn if wedge elements with full integration were used.

The 4 scenarios have also been studied under uniaxial tensile strain (restricted Poisson) and shear load (with free and restricted Poisson). In the uniaxial tensile strain case, the results and conclusions are equivalent to the ones displayed in Fig. 9. The only difference is that in the case of restricted Poisson, the stiffness is higher and the final strain is reached earlier than in the uniaxial tension stress (free Poisson). Readers should be warned about the convergence issues of this load condition, which can lead to unexpected results. For example, this load condition in the scenario SC-ABS* with user defined tolerance (10^{-6}) stops at $\epsilon_{yy} < 0.05$ while SC-REL* with the same user defined tolerance reaches the expected solution like the N2N scenarios. The optimization of the user defined tolerance is out of the scope of this paper. The simulations have been done with the same user defined tolerance as in the uniaxial stress load with free Poisson. On the other hand, Fig. 10 shows the homogenized stress-strain behavior for the 4 scenarios under shear load with free Poisson effect. There are small differences between the node-to-node and surface-coupling implementations. Note that the surface-coupling simulations do not reach the imposed macro-strain ($\epsilon_{xz}^0 = 0.12$). They stop sooner due to convergence problems, see Fig. 10. These differences are again due to the solver tolerance limit based on the average force. Using a user defined limit based on the average force of 10^{-6} , the surface-coupling implementations give the same results as node-to-node ones, the curve SC-ABS* in Fig. 10

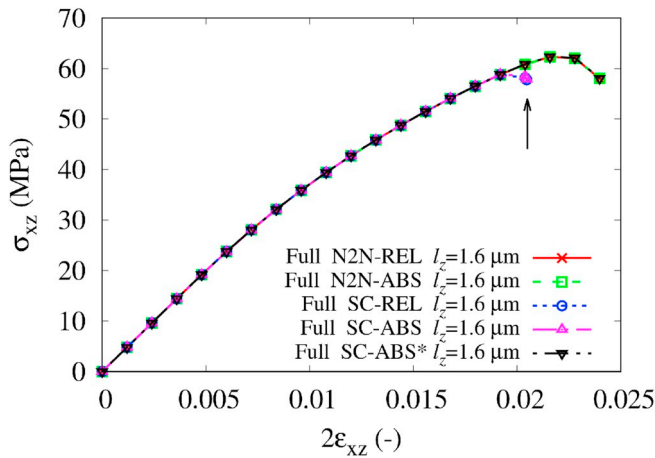


Fig. 10. Homogenized stress-strain response for the four different implementations under in-plane shear load with free Poisson. RVE with $l_z = 1.6 \mu\text{m}$ discretized with full integration hexahedral dominated mesh. SC-ABS* surface-coupling implementation with user defined tolerance (10^{-6}).

overlaps the N2N curves. Finally, the results under shear load with restricted Poisson are equivalent to the ones with shear load and free Poisson.

Table 8 shows the CPU time given by the solver for the simulations with different scenarios and four load cases. The RVE model with $l_z = 1.6 \mu\text{m}$ discretized with full integration hexahedral mesh has been used in all cases. All the simulations were done in a personal computer with an Intel CORE™ i7 processor and 16 GB of RAM. Although parallelization with multiple processors has been used to decrease the wall clock time, the CPU time takes into account the time spent in all the CPUs. It can be observed that for all the load cases, the N2N scenarios spend more CPU time than the SC scenarios without user defined control. This is expected because SC scenarios have a fewer restrictive tolerance, and therefore they reach the convergence in each step with fewer iterations. When the user defined tolerance is given, the CPU time of the SC scenarios increases. It is noteworthy that smaller tolerance limit leads to a larger increase of CPU time. Therefore, the tolerance limit must be handled with care to reach an accurate result without greatly increasing the computational cost.

Looking at the uniaxial load with restricted Poisson in Table 8, the N2N scenarios spend more CPU time than the uniaxial load with free Poisson. As commented before, the SC scenarios under this load have convergence issues and this is reflected in the CPU times for SC scenarios. On the other hand, the CPU times under shear load with free and restricted Poisson (Table 8) are smaller than the uniaxial cases because there are fewer convergence issues due to the material models, matrix plasticity and debonding. Again, decreasing the user defined tolerance improves the solution but increases the CPU time.

As a conclusion, when the user defined tolerance is given, there are no significant differences in the stress-strain behavior between the studied implementations of PBCs using the implicit solver Abaqus/Standard [36]. In addition, the results reported in sections 5.3 and 5.4 using different depths and element types can be reproduced with all the

studied implementations of Table 6.

6. Results using PBCs under dynamic load (explicit solver)

The finite element model with PBCs described in section 5 has been studied under dynamic loads using an explicit solver, Abaqus/Explicit [36]. The loads are introduced in the same manner as for the implicit solver by imposing the displacements on the dummy nodes. These displacements are set to increase linearly with the time of the simulation. Consequently, the dummy nodes are moved at a constant velocity and materials are subjected to a constant strain-rate.

To allow a comparison of the dynamic results with the previous results of static loads, two hypotheses have been done: (i) the material properties of fiber, matrix and interface were not changes (strain-rate independent); and (ii) the simulation time is large enough to relax the stress waves created at the beginning, see section 6.1. Under these conditions, the explicit method can be used to mimic a quasi-static process. One advantage of the explicit solver is that it can easily handle non-linearities, avoiding convergence issues due to the material models.

Because the material is strain-rate independent, a high strain-rate of 100 s^{-1} has been imposed as a dynamic load to reduce the simulation time and CPU time. In this section, results with different RVE depths l_z , element types, and implementations of PBCs have been compared under uniaxial and in-plane shear loads. The default parameters of the explicit solver were used. Finally, it is worth mentioning that double precision has been used in all the simulations.

6.1. Stress waves and mass-scaling

The explicit solver is based on the calculation of nodal acceleration for a given time step. Therefore, stress waves are propagated through the elements. Because PBCs introduce displacements at the surface of the RVE, the stress waves are initially generated at these surfaces traveling to the center. After an initial relaxation time t_r , the stress waves and reflections have damped out over the whole volume of the RVE, leading to a quasi-static stress state until the end of the simulation.

The speed of sound can be estimated for each component of the composite using the isotropic formulation $v_s = \sqrt{E/\rho}$. This speed in the matrix (1520 m/s) is lower than the speed in the transverse direction of the fiber (2970 m/s). The speed of sound in the matrix can be used to obtain the order of the relaxation time, which is defined as the time needed for a wave to cross the transverse length of the RVE. Using a rounded value of the RVE transverse size of $12 \mu\text{m}$, the $t_r \approx 7 \text{ ns}$. On the other hand, the total simulation time has been calculated imposing a high strain-rate of 100 s^{-1} . The maximum strain under uniaxial tension and shear is 0.01 and 0.012, respectively, therefore the corresponding simulated times are $100 \mu\text{s}$ and $120 \mu\text{s}$. The quasi-static behavior is reached in the early part of the simulation ($t_r \approx 7 \text{ ns}$).

Another aspect to be considered is that the maximum time step to perform a stable explicit analysis must be less than the time needed for a sound wave to travel across a single element, i.e. the Courant-Friedrichs-Lewy condition [45]. For the global element size of $0.4 \mu\text{m}$, the simulation time becomes very large due to the high number of iterations computed by the solver ($\approx 10^7$). This high number of iterations leads to expensive simulations with hours of CPU time for small

Table 8

CPU time in minutes for the static simulations of each scenario under four load cases. RVE model $l_z = 1.6 \mu\text{m}$, hexahedral dominated mesh with full integration elements. SC-REL* and SC-ABS* are the surface-coupling scenarios with user defined tolerance.

Load	N2N-REL	N2N-ABS	SC-REL	SC-ABS	SC-REL*	SC-ABS*
Uniaxial free Poisson	71.2	73.7	16.9	13.52	89.04	89.04
Uniaxial restricted Poisson	118.1	120.9	1.5 ^a	1.0 ^a	37.5	26.9 ^a
Shear free Poisson	5.4	5.4	3.03	1.6	8.24	3.6
Shear restricted Poisson	4.2	4.1	2.3	2.3	4.2	4.2

^a The simulation stops earlier than expected due to convergence issues.

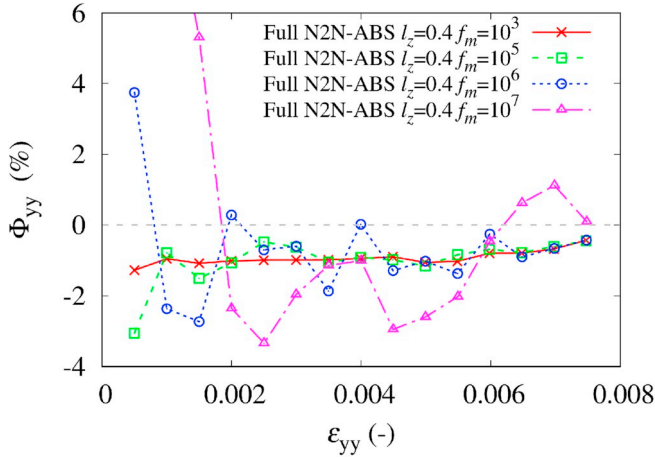


Fig. 11. Difference in vertical homogenized stress (σ_{yy}) between explicit and implicit solvers (7) as a function of the homogenized strain. RVE model $l_z = 0.4 \mu\text{m}$ with N2N-ABS implementation discretized with full hexahedral integration elements. Results increasing $f_m =$ mass factor.

models like the selected RVE, and they can also contain inadmissible round-off errors.

In order to speed up the simulations, the mass of the system has been scaled by increasing the density of the components. When the density is multiplied by a mass factor $f_m > 1$, the speed of sound decreases and the minimum time step increases by $\sqrt{f_m}$. A larger time step leads to a smaller number of iterations. Because the speed of sound decreases, the t_r must be also recalculated. For example, using a $f_m = 10^5$ gives a $t_r \approx 2.5 \mu\text{s}$ which is still lower than the simulation time $100 \mu\text{s}$.

Simulations under uniaxial load with free Poisson have been performed with the N2N-ABS implementation. The RVE with 4 fibers and $l_z = 0.4 \mu\text{m}$ has been discretized with full integration hexahedral dominated mesh. Fig. 11 shows the difference in the percentage of the vertical homogenized stress σ_{yy} between the explicit results and the reference behavior obtained with the implicit solver.

$$\Phi_{yy} = 100 \left(\frac{\sigma_{yy}^{\text{explicit}} - \sigma_{yy}^{\text{implicit}}}{\sigma_{yy}^{\text{implicit}}} \right) \quad (7)$$

The four curves show the results obtained using different mass factors, $f_m = 10^3, 10^5, 10^6$ and 10^7 . When the f_m is larger than 10^5 , oscillating results are distinguished along the whole simulation. This means that the t_r is comparable to the simulation time, and therefore no quasi-static solution is reached. When the f_m is 10^5 , the initial oscillations become insignificant for $\varepsilon_{yy} > 0.004$. When the $f_m < 10^5$, the initial oscillations are hidden before the first extracted data point, $\varepsilon_{yy} = 0.005$. It is noteworthy that the explicit results ($f_m \leq 10^5$) have a constant deviation from the implicit results ($\Phi_{yy} = 0$) about 0.5 MPa. Additionally, the Von Mises stress field of the explicit simulation with $f_m = 10^5$ has been compared with the stress field calculated with the implicit solver. The average of the absolute difference between the stresses in each solid element is lower than 0.6 MPa for simulation times larger than the relaxation time. Therefore, a mass factor of 10^5 has been chosen to speed up the simulation in the following simulations.

It is worth mentioning that a different explicit solver might have a different mass-scaling implementation or strategy. Nevertheless, it is the researcher's responsibility to check it in order to guarantee that the material response remains fully unaltered when mass-scaling is applied.

6.2. Depth of RVE

Fig. 12 shows the homogenized stress-strain curves for three RVEs with different depth, $l_z = 0.4, 1.6$ and $3.2 \mu\text{m}$, which correspond to 1, 4

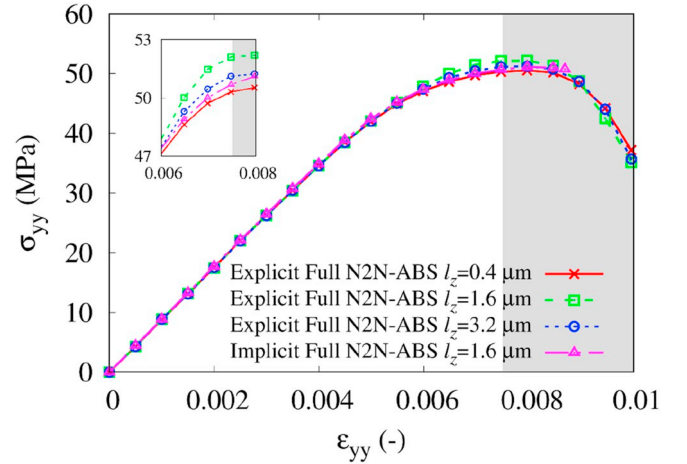


Fig. 12. Homogenized stress-strain response for different RVE depths using an explicit solver with $f_m = 10^5$. As a reference, the homogenized response of the RVE model $l_z = 1.6 \mu\text{m}$ using the implicit solver has been added. N2N-ABS implementation discretized with full integration hexahedral dominated mesh.

and 8 elements through the depth, respectively. As a reference, the homogenized result using the implicit solver is included. The uniaxial tensile stress with free Poisson has been imposed in the vertical direction y . The N2N-ABS implementation has been used with the full integration hexahedral dominated mesh. Small differences ($< 2 \text{ MPa}$) between the curves with different depth are visible after the linear behavior when matrix plasticity and debonding start to be active (see inset Fig. 12). Results with homogenized strain beyond 0.75% have been discarded due to convergence issues in the implicit solver (see section 5.3). The PBCs combined with the material model (brittle debond) lead to a local peak in stress at the vertices which increases when the RVE depth increases, as is in the case with the implicit solver 5.3. In addition, these local stresses generate dynamic waves that affect the propagation of debonds, thereby modifying the maximum homogenized strength. Unlike the implicit case, there are no convergence issues and the simulations reach the maximum strain imposed by the PBCs. The results obtained with the explicit solver after the maximum stress must be validated with experiments to check the viability of the chosen materials models when PBCs are applied.

When the applied load is a uniaxial tensile strain (restricted Poisson), the results with different depths are equivalent to the ones in the case of uniaxial tensile stress (free Poisson). Therefore, small differences in the homogenized stress are observed when the RVE depth increases. On the contrary, when the shear load is applied, there are no relevant differences between RVEs with different depths.

A RVE with one element through the thickness ($l_z = 0.4 \mu\text{m}$) gives the closest results to the simulations with the implicit solver, see Fig. 12. Therefore, only one element through the thickness is used in the following simulations with the explicit solver.

6.3. Element type

Simulations have been performed under different load conditions, uniaxial and in-plane shear, using hexahedral elements with full (C3D8) [36] and reduced integration (C3D8R), as well as wedge elements with reduced integration (C3D6R). Three meshes have been compared: hexahedral dominated mesh with full integration elements; hexahedral dominated mesh with reduced integration elements; and, wedge mesh with reduced integration elements. Note that the explicit solver, Abaqus/Explicit [36], only has wedge elements with reduced integration.

The explicit solver gives similar results as the implicit solver when full and reduced integration meshes are compared, see 5.4. There is

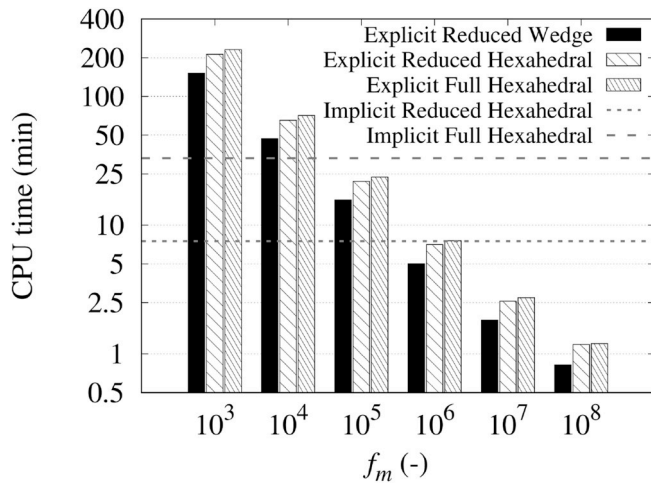


Fig. 13. The bars represent the CPU time of the explicit simulations versus the mass-scaling factor f_m for three different meshes: full integration hexahedral dominated mesh; reduced integration hexahedral dominated mesh; and reduced integration wedge mesh. The dashed lines represent CPU time of the implicit simulations for two meshes: full integration hexahedral dominated mesh; and reduced integration hexahedral dominated mesh. Note that both axes are in logarithmic scale.

hourglassing when elements with reduced formulations are used in a RVE with several layers through the depth under uniaxial loads. The hourglassing is limited if the RVE only has one layer of elements through the depth, or if shear loads are applied.

In order to compare the CPU time of each mesh, the RVE model with one layer through the depth $l_z = 0.4 \mu\text{m}$ has been selected. The uniaxial load condition with free Poisson is applied to the model with N2N-ABS implementation. All the simulations have been done without parallelization because the RVE model with one element layer through the depth and PBCs does not allow automatic parallelization of the explicit solver [36]. However, automatic parallelization is possible using RVE models with more than one layer of elements through the depth and any proposed implementation of PBCs. The bars in Fig. 13 shows the CPU time of the explicit simulations with the three different meshes and various amounts of the mass-scaling factor, f_m . The CPU time decreases when the f_m increases for all the meshes. There is a linear relationship between the logarithm of the CPU time and the logarithm of the f_m . The CPU time decreases only 8% when reduced hexahedral elements are used instead of full hexahedral elements for the studied range of f_m . This decrease is much lower than the expected value of 77% as in the implicit solver. It has been checked that using different boundary conditions in the explicit solver, such as a uniform displacement of one surface, the CPU time between full and reduced integration elements decreases as expected (77%). Therefore, for the studied explicit solver the implementation of PBCs is a limitation when reduced integration elements are used. On the other hand, using reduced wedge elements decreases the CPU time by 34%, which is the same decrease obtained for the implicit solver. It can be concluded that simulations with reduced hexahedral elements are less attractive in the explicit solver because they introduce hourglassing without decreasing too much the CPU time.

Additionally, the CPU times of implicit simulations with reduced and full hexahedral elements have been included in Fig. 13, see the dashed lines at 7.5 and 33.1 min, respectively. The CPU times of the explicit simulations with full integration hexahedral element are below 33.1 min when the $f_m \geq 10^5$. In the case of reduced integration hexahedral elements, the CPU times of the explicit simulations are only below 7.5 min when the $f_m \geq 10^6$. It is noteworthy that at this mass-scaling the stress waves are relevant during the whole simulations and affect the results.

Table 9

CPU time in minutes for the explicit simulations of each scenario under different loads. RVE model $l_z = 0.4 \mu\text{m}$, hexahedral dominated mesh with full integration elements.

Load	N2N-REL	N2N-ABS	SC-REL	SC-ABS
Uniaxial free Poisson	14.5	22.4	>1440 ^a	6.3
Uniaxial restricted Poisson	14.5	22.2	4.7	5.6
Shear free Poisson	17.6	27.0	>1440 ^a	7.6
Shear restricted Poisson	17.2	26.9	5.8	6.9

^a The simulations have only reached 10 % of the imposed macro-strain after 24 h.

Judging from the above, a RVE with one layer of elements through the depth and full integration elements is the most favorable model to perform simulations in the explicit solver under different loads.

6.4. Results with different implementations

Results with different implementations of PBCs (Table 6) are compared in this section using the explicit solver Abaqus/Explicit [36]. The RVE with 4 fibers and depth $l_z = 0.4 \mu\text{m}$ has been discretized with full integration hexahedral dominated mesh. Uniaxial and in-plane shear loads have been applied with a fixed mass factor, $f_m = 10^5$. Note that the copied nodes created in the surface-coupling scenarios do not belong to any solid element, see copied nodes (x-) in Fig. 2. Therefore, they do not have any mass. In order to perform the explicit analysis, the average mass value of the nodes that belong to the solid elements has applied to them.

There are no significant differences between the homogenized stress-strain results using the different scenarios (Table 6). There are also no differences between the stress fields when they are compared for each time step. On the contrary, the CPU time to perform the same simulation can change abruptly between the different scenarios. Table 9 shows the CPU time in minutes of each implementation for the different load cases. The N2N-ABS implementation takes more CPU time than the N2N-REL in all the studied loads. Note that the CPU time to solve the load with free or restricted Poisson is equal in the N2N implementations. On the other hand, the CPU time for the SC implementations is much smaller than the N2N implementations in the two load cases with restricted Poisson. However, for the load cases with free Poisson, the SC-REL requires an exceptional long CPU time while SC-ABS remains faster than N2N implementations. When there is more than one layer of elements through the depth, the CPU time for the SC-ABS also increases to values similar to those for SC-REL. A discussion with Abaqus developers suggests that surpassing a certain number of equations can trigger extra algorithms to perform the simulations, causing a drastic increase in CPU time. In the current study, the element size is fixed, therefore increasing the depth of the RVE is equivalent to increasing the number of equations.

It can be concluded that it is advantageous to use the SC implementations when the load cases have restricted Poisson effect for conformal or non-conformal meshes with different depths of the studied 4-fibers RVE. Note that the number of equations increases with the depth. On the contrary, for loads with free Poisson effect, the N2N-REL implementation for conformal meshes gives better performance. For the case of non-conformal mesh and load with free Poisson, the SC-ABS implementation can be used if the RVE has only one layer of elements in the studied 4-fibers RVE (limited number of equations). If this latter case does not work properly, it is recommended to move to a conformal mesh and use N2N-REL implementation.

6.5. Comparison between implicit and explicit solver

Finally, the results obtained using the explicit solver have been compared with the results from the implicit solver. Fig. 14 shows the

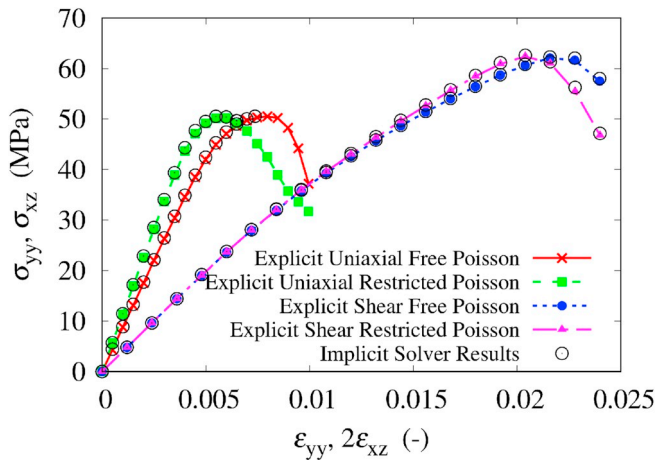


Fig. 14. Stress-strain curves for different implementations under uniaxial and shear load using an explicit solver with $f_m = 10^5$. The results from the implicit solver are also included with black circles. RVE model $l_z = 0.4 \mu\text{m}$ discretized with full integration hexahedral dominated mesh using N2N-ABS implementation.

homogenized stress-strain curves for uniaxial and shear load cases with free and restricted Poisson. Geometry, mesh, and implementation of the studied model are: RVE with 4 fibers and $l_z = 0.4 \mu\text{m}$, hexahedral elements with full integration; and N2N-ABS. The lines represent the material behavior calculated with the explicit solver while the results from the implicit solver are represented by black circles. The results are in good agreement for the linear and non-linear response. It is noteworthy that all the simulations performed with the explicit solver reach the imposed macro-strain without convergence issues. Therefore, material models with damage can be used in a larger range of strains.

7. Conclusions

A systematic study of different implementations of PBCs in a finite element model has been performed for static and dynamic loads for which implicit and explicit solvers have been used, respectively. A unidirectional fiber composite has been modeled with a Representative Volume Element (RVE). The material model of the matrix includes plasticity and damage, and the interfaces between fibers and matrix have been modeled with cohesive surfaces to allow debonding. The implementations of PBCs are compatible with damage propagation through opposite surfaces of the RVE for conformal and non-conformal meshes. The RVE has been subjected to uniaxial and in-plane shear loads with free and restricted Poisson effect. The homogenized stress-strain results and CPU times have been compared for the different implementations of PBCs. In addition, the most relevant parameters such as the RVE depth and element type have been identified and discussed.

Using the implicit solver, high local stresses at the vertices of the RVE and convergence issues have been found during the propagation of debonding under uniaxial loads. These results are a consequence of the combination of the geometry, material models, load case and PBCs. The high local stresses at the vertices become relevant when the depth of the RVE increases, although no significant differences have been observed in the homogenized stress-strain curves. Elements with reduced integration lead to hourglassing which is more severe when the RVE depth increases. Although all the implementations of PBCs are mathematically equivalent, the results of the homogenized stress-strain behavior and CPU time show some differences between node-to-node and surface-coupling implementations. These differences have been identified and corrected by using the same tolerance limit in all the implementations.

Quasi-static simulations with the explicit solver need mass-scaling

to decrease the CPU time. To this end, the density of both materials, fiber and matrix, has been increased. It has been found that for the studied RVE model, a mass factor of 10^5 reduces the CPU time considerably, while the homogenized results still have small differences with the reference solution obtained with the implicit solver. As in the implicit solver, high local stresses at the vertices are present when the depth of the RVE increases under uniaxial loads. If reduced integration elements are used, hourglassing is significant in a RVE with more than one layer through the depth under uniaxial load. The reduction of CPU time using reduced integration elements is not as relevant as in the simulation with an implicit solver. Finally, no significant differences in the homogenized stress have been observed between the different implementations of PBCs. However, there are relevant differences in the CPU time. For loads with free Poisson, the N2N-REL is generally the fastest implementation while for loads with restricted Poisson the SC-REL is the fastest one.

The results of the quasi-static simulation obtained from the implicit and explicit solvers have been compared with good agreement. Nevertheless, it is important to know the general behavior and limits of the solver to select the best implementation of PBCs. As a conclusion, none of the scenarios of PBCs is consistently better in the studied solvers. Some useful notes are described to avoid unexpected results using PBCs:

- Using node-to-node implementations in an *implicit solver* can avoid incompatibilities between the damage material models (or cohesive surfaces) and the surface-coupling implementation where the copied nodes are tied to the nodes of the RVE surface.
- When a model with non-conformal mesh is solved with the surface-coupling implementation in an *implicit solver*, the results must be verified considering possible convergence issues due to tolerance limits.
- If the RVE model has more than one layer of elements through the depth, full integration elements are strongly recommended to avoid hourglassing when matrix plasticity and debonding are relevant in *implicit and explicit solvers*.
- When mass-scaling is used to reduce the CPU time in an *explicit solver*, it is important to verify that the stresses reach the quasi-static regime.
- Surface-coupling implementations can speed up the simulations for conformal and non-conformal meshes under loads with restricted Poisson in an *explicit solver*.
- Be careful when loads with free Poisson are applied. These loads can introduce extra conditions in an *explicit solver* increasing drastically the CPU time.

Acknowledgments

The work leading to this publication has been funded by the SBO project “M3Strength” (Grant no. 130546), which fits in the MacroModelMat (M3) research program, coordinated by Siemens (Siemens PLM Software, Belgium), funded by SIM (Strategic Initiative Materials in Flanders) and VLAIO (Flemish government agency Flanders Innovation & Entrepreneurship).

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